Hyperspectral Band Selection via Difference Between Intergroups

Shuying Li, Baidong Peng, Long Fang, Qiang Zhang, Lei Cheng, and Qiang Li, *Member, IEEE*

***Abstract*— Various methods are proposed to reduce the dimensions of hyperspectral image (HSI) by band selection in recent years. Most methods select one band from each group to construct a band subset. However, the redundancy in the selected bands from different groups is neglected. Furthermore, the researchers do not pay enough attention to how many bands are appropriated for selection. To solve these issues, we propose a hyperspectral band selection method via difference between intergroups (DIG), which includes grouping strategy and ranking strategy. Specifically, the grouping strategy adopts intragroup similarity to reasonably distribute all partitioning point positions. The similarity of bands within the same group is significantly improved. For the ranking strategy, it not only takes into account the knowledge and intragroup similarity of bands, but also evaluates the differences between each band and other intergroup bands. The redundancy in band subset is reduced sufficiently. To accurately obtain the optimal number of bands, an evaluation function is designed to measure the information content and redundancy in various band subsets. Experimental results from different aspects show that the proposed model has a large performance advantage on three public datasets.**

***Index Terms*— Band selection, difference between intergroups (DIG), hyperspectral image (HSI), optimal number of bands.**

I. Introduction

HYPERSPECTRAL image (HSI) [1] is composed by dozens to hundreds of bands. Due to HSI having affluent ground object information [2], it is successfully introduced into target detection [3], biological analysis [4], precision agriculture [5], and so on. However, the large number of bands lead to the increase in HSI dimension [6]. Moreover, the high similarity between adjacent bands causes data redundancy [7], [8], [9], [10]. To address these troubles, dimensionality reduction is required for HSI.

Feature selection (band selection) [11], [12], [13], [14], [15] and feature extraction [16], [17], [18], [19], [20] are normally employed in HSI dimensionality reduction. Feature extraction aims to project the high-dimensional data of HSI into low-dimensional space and extract the main feature information. In contrast to feature extraction, feature selection can physically preserve the messages in the original data. Currently, band selection is used in most methods to reduce the HSI dimension, which aims to choose as few bands as possible to achieve all bands functionality. According to whether the samples are effectively utilized, band selection methods are divided into supervised [21], [22], semi-supervised [23], [24], [25], [26], and unsupervised [11], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40] methods. In practice, it is expensive to label samples. Considering the problem, this article mainly focuses on unsupervised methods. At present, there are two main difficulties associated with band selection. One is to choose the band subset with high information content and low redundancy. Another difficulty is to accurately acquire the most appropriate number of bands.

For the first difficulty, various approaches are proposed, which mainly include clustering, ranking, and greedy methods. The clustering methods [31], [32] take higher similar bands into the same cluster. All cluster centers are chosen to construct band subset. However, in these methods, the data content is neglected. Different from clustering methods, ranking methods [33], [34], [35] measure and rank the physical data of bands. Various bands with larger weights are chosen to build subset. Nevertheless, these methods ignore the redundancy in band subset, which inevitably lead to the poor performance. With regard to the greedy methods [11], [36], [37], they build a candidate band subset and adopt traversal manner to add or remove bands. Moreover, they design objective functions to measure the performance of band subset. Unfortunately, the traversal manner inevitably results in a large amount of computation. In recent years, researchers adopt the combination based on clustering and ranking methods. These methods [38], [39], [40] have strong advantages compared to traditional clustering and ranking methods. Based on the band clustering, the band subset is constructed by choosing the maximum ranked band in each cluster. Nevertheless, these methods ignore the correlations in different groups. The band subset may contain a certain amount of redundancy, and there is still room for further improvement.

The difficulty in optimal number is not received as much attention as the first difficulty. To cope with this problem, a model based on band selection is designed by the curve figure [37], [41]. The number of bands and classification accuracy are the horizontal and vertical coordinates of the figure. The inflection point is the optimal number of bands. Clearly, these approaches require obtaining the classification

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accuracy with various band subsets. Classification accuracy is affected by the classifier, which is severely unstable.

To address these problems, we propose a hyperspectral band selection method via difference between intergroups (DIG). DIG fully combines the advantages of the grouping strategy and the ranking strategy. The grouping strategy aims to make the bands more dispersed and reduce the redundancy in the bands of different groups. The ranking strategy combines intragroup and intergroup information, which makes the band subset more representativeness. Based on the above two strategies, we propose a method to decide how many bands are most appropriate. The major contributions are as follows.

1) The grouping strategy via intragroup similarity (GSIS) is proposed to reduce the similarity between various groups. Different from other criteria, the intragroup similarity is designed by all the intragroup bands to judge the rationality within the groups, which adequately ensures that the bands are more dispersed.
2) Based on the GSIS, the ranking strategy via difference between intergroups (RSDI) is designed, which not only takes advantage of the knowledge and similarity in each group, but also considers the difference in diverse groups. The redundancy and the performance in the band subset are reduced and further improved.
3) An automatic method is proposed to determine the optimal number of bands by the evaluation function, which can produces multiple band subsets and get the inflection point in one operation. This evaluation function can accurately measure the performance of band subset by the knowledge and redundancy.

The rest of this article is organized as follows. Section II introduces several unsupervised approaches. Section III describes the proposed grouping strategy, ranking strategy, and the optimal number method. Section IV compares the performance with other approaches on three public datasets. Section V summarizes the proposed method.

II. Related Work

As described in Section I, the approaches based on band selection can be divided into clustering methods, ranking methods, and greedy methods. The representative methods are introduced in this section.

A. Clustering Methods

The clustering methods build the similarity matrix and design the objective function to optimize the clusters. Classical clustering methods include affinity propagation (AP) [31], hyperspectral band selection via an adaptive subspace partition strategy (ASPS) [39], fast neighborhood grouping method for hyperspectral band selection (FNGBS) [40], and so on. AP constructs the similarity matrix by K-L divergence and Gaussian peak. After obtaining the similarity matrix, the attractiveness in each band is adopted to evaluate its importance. ASPS and FNGBS combine the grouping strategy and ranking strategy. For the convenience of description, these two methods are also classified as clustering methods. ASPS sets the best location of the partitioning points by maximizing the ratio of the intergroup distance to the intragroup distance in the neighborhood subspace. Afterward, the band noise is applied as an evaluation metric to construct band subset. Different from ASPS, FNGBS assumes that the undetermined region is the area between the adjacent subspace centers. The label of each band in the region is set as the subspace center with the minimum difference. The location in the new partitioning point is where the labels of these bands change. Over several iterations, the variation of the subspace gradually stabilizes. Furthermore, the local density and information entropy are utilized to obtain the most appropriate band in each subspace. These methods consider the correlation information between bands, which ensures that the chosen bands are sufficiently. Nevertheless, the redundancy in these bands from different groups is ignored.

B. Ranking Methods

Ranking methods normally employ statistics (information entropy, mutual information, K-L divergence) to rank all bands. Subsequently, the bands are arranged in ascending or descending order and the corresponding bands are acquired as band subset. Typical ranking methods include enhanced fast density peak based clustering (E-FDPC) [33], maximum variance principal components analysis (MVPCA) [34], linearly constrained minimum variance based constrained band selection (LCMV-CBS) [35], and so on. E-FDPC calculates and normalizes the local density and intragroup distance. Whereafter, the exponential based learning rule is adopted to compute the cutoff threshold. The local density and intragroup distance are designed to choose bands. Similar to E-FDPC, MVPCA computes the Euclidean distance in diverse bands to construct the similarity matrix. The loading factor matrix is constituted by decomposing the eigenvalues in the similarity matrix, which is utilized to rank the bands. Unlike the other two methods, LCMV-CBS designs a filter to measure the similarity between each band and the others. Afterward, the filter ranks bands in descending order and chooses the front bands. Obviously, these methods only focus on the information content in the bands and neglect the similarity between selected bands. This fashion easily chooses the bands with extreme similarity, which could result in a poor performance.

C. Greedy Methods

Greedy methods employ a search strategy to acquire the band subset with the best overall performance. First, the band subset is updated in a specific order. The discriminant function decides whether the corresponding update is valid or not. The main greedy methods are orthogonal projection-based band selection (OPBS) [36], volume gradient band selection (VGBS) [11], sequential forward selection (SFS) [37], and so on. OPBS chooses the band with the maximum variance as the initial band subset. Afterward, the OP vector for the other bands is calculated. The band with the maximum OP vector is added to the band subset. OPBS stops iterating when the number of selected bands reaches a specified number. VGBS takes all bands as candidate bands, and these bands form a parallelotope. Subsequently, the band is removed if it causes

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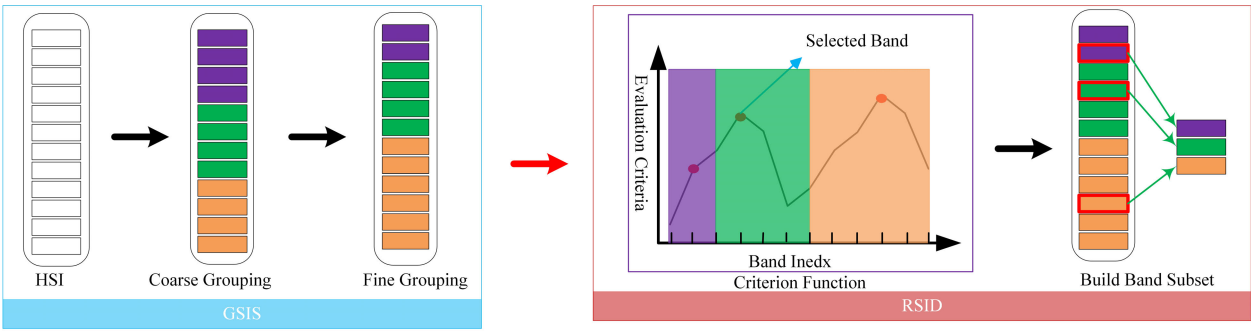


Fig. 1. Flowchart of DIG, including GSIS and RSDI. GSIS aims to reduce the similarity between different groups by dual-stage grouping. Based on the multiple groups constructed by GSIS, RSDI uses among information entropy, intergroup, and intragroup information to construct evaluation criteria. The evaluation criteria are used to select the most representative band.

the minimum volume loss in the parallelotope. In contrast to VGBS, SFS increments bands from an empty band subset until the number reaches the specified requirement. These methods employ search strategies to reach the optimal solution from a global perspective, which results in a large time cost. Furthermore, due to the characteristics of the HSI data, they can only produce a locally optimal solution using a suboptimal search strategy.

III. Proposed Method

The proposed approach is introduced in this section. The flowchart mainly contains GSIS and RSDI, as shown in Fig. 1. Specifically, GSIS decides the partitioning point locations by the intragroup similarity. According to the generated several groups, RSDI calculates the product of the information entropy, the local density in intragroup, and the DIG to choose representative band. Based on the GSIS and RSDI, we design an evaluation function to obtain the optimal number of bands.

A. Grouping Strategy via Intragroup Similarity

Adjacent bands exhibit higher similarity than non-adjacent bands, which is the major difference between HSI and other data. Based on higher similarity between adjacent bands, we propose a coarse-fine grouping strategy. First, coarse grouping divides the bands into several groups on average based on their index. It forms various partitioning points between adjacent groups. Subsequently, fine grouping moves these partitioning points by intragroup similarity, changing the number of bands in each group.

1) Coarse Grouping: Suppose $X \in \mathbb{R}^{N\times L} = \{X_i\}_{i=1}^{i=L}$ is HSI, where N and L denote the pixels and band numbers, and X_i is the ith band. The band index is ordered and sequential. To accelerate the running time, coarse grouping is to equally divide all bands into m groups. The coarse grouping aims to equally divide all bands into m groups. For m groups, since the order of bands is not changed in our grouping strategy, the $m+1$ partitioning point positions can more clearly describe the bands contained in each group. The partitioning point P_g is designed as follows:

$$P_g = \begin{cases} 1, & g = 1 \\ \left\lfloor \dfrac{L \times g}{m+1} \right\rfloor, & g = 2, 3, \ldots, m \\ L, & g = m+1. \end{cases} \tag{1}$$

Q_g is the gth group. Furthermore, the bands in Q_g is between P_g and P_{g+1}. Q_g is established with

$$Q_g = \begin{cases} \{X_i\}_{i=P_g}^{i=P_{g+1}-1}, & g = 1, 2, \ldots, m-1 \\ \{X_i\}_{i=P_m}^{i=P_{m+1}}, & g = m. \end{cases} \tag{2}$$

2) Fine Grouping: The coarse grouping strategy simply partitions by band order, which lacks theoretical justification. The bands with extreme similarity may still be in different groups. Therefore, the fine grouping strategy is adopted to achieve the ideal result. The traditional fine grouping approaches [31], [40] decides the partitioning points by comparing the similarity between the undetermined band and group center. However, the effects of bands in other intragroup are neglected. To address the issue, the intragroup similarity in the neighborhood group is applied to move the partitioning points. The intragroup similarity is the degree of resemblance between all intragroup bands and the group center. The group center σ_g in Q_g is

$$\sigma_g = \left\lfloor \frac{P_{g+1} - P_g}{2} \right\rfloor + P_g. \tag{3}$$

To acquire the ideal partitioning point position, we first calculate the Euclidean distance with all bands to build the similarity matrix S. The movement of a partitioning point can affect the intragroup similarity in the left and right groups. For the neighborhood group established from Q_g and Q_{g+1}, we assume that the intragroup similarity V_g^i is the average Euclidean distance between all bands and the group center when X_i is the partitioning point, which is designed as follows:

$$V_g^i = \frac{\sum_{j=P_g}^{i-1} S_{ij}}{i - P_g} + \frac{\sum_{j=i}^{P_{g+2}} S_{ij}}{P_{g+2} - i} \tag{4}$$

where S_{ij} denotes the Euclidean distance between X_i and X_j. The smaller value of the intragroup similarity, the more reasonable the location of the partitioning point. First, when P_{g+1}, $P_{g+1}-1$ or $P_{g+1}+1$ is partitioning point, the values of intragroup similarity are generated by (4). If the value of $P_{g+1}-1$ or $P_{g+1}+1$ is the minimum result, the movement direction for P_{g+1} is left or right, respectively. We assume that the undetermined region of the partitioning point is between P_{g+1} and the group center corresponding to the movement direction. The intragroup similarity of all bands in this region is generated when they are partitioning point. The band with

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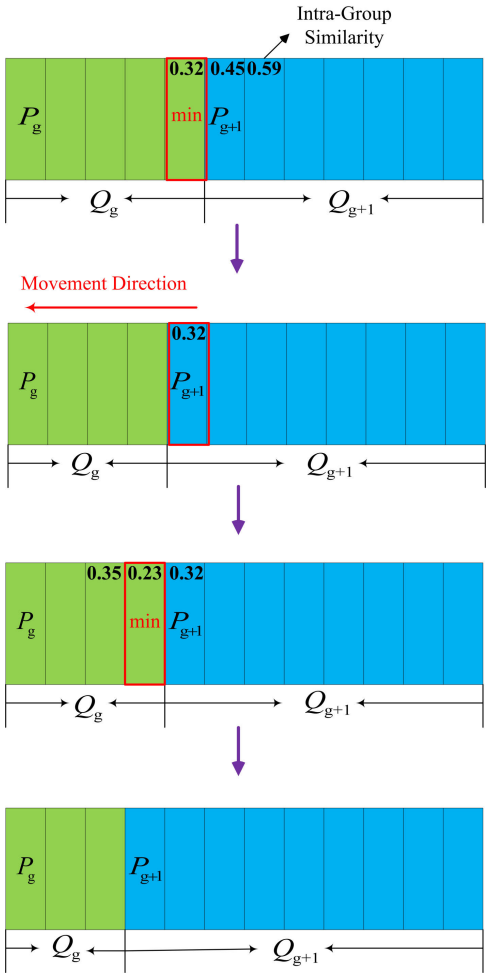


Fig. 2. Assume that the input and output are produced by coarse grouping and fine grouping, respectively.

the minimum value is replaced as P_{g+1}. An example for the fine grouping in neighborhood groups Q_g and Q_{g+1} is described in Fig. 2. A more detailed explanation for the fine grouping is shown in Algorithm 1. After updating all partition points, $\{Q_g\}_{g=1}^{m}$ is rebuilt by (2).

B. Ranking Strategy via Difference Between Intergroups

Recent band selection methods [38], [39], [40] only choose one representative band in each group. This fashion may lead to certain redundancy in band subset. To address the issue, we design a ranking strategy that combines local density in intragroup, DIG and information entropy, which aims to choose the band with the largest information amount and the smallest redundancy within each group. These three factors are utilized to measure intragroup similarity, intergroup dissimilarity and information content for each band, respectively. The band with the maximum weight in each group is chosen to establish the band subset. These three factors are described in the following.

Local density is calculated by the number of bands in a certain range, which is the ratio between this number and the total number of bands. Here, this deterministic range is referred to as the threshold. To obtain intragroup similarity for

Algorithm 1 Fine Grouping

Input: Coarse grouping results of partitioning points $\{P_g\}_{g=1}^{g=m+1}$.

Output: Fine grouping results of partitioning points $\{P_g\}_{g=1}^{g=m+1}$.

```
for g = 1 to m − 1 do
    σ_g = ⌊(P_{g+1} − P_g)/2⌋ + P_g;
    σ_{g+1} = ⌊(P_{g+2} − P_{g+1})/2⌋ + P_{g+1};
    if V_g^{P_{g+1}−1} < V_g^{P_{g+1}+1} & V_g^{P_{g+1}−1} < V_g^{P_{g+1}} then
        for i = σ_g to P_{g+1} − 1 do
            P_{g+1} = P_{g+1} − 1;
            if V_g^i < V_g^{P_{g+1}} then
                P_{g+1} = i;
            end
        end
    end
    if V_g^{P_{g+1}+1} < V_g^{P_{g+1}−1} & V_g^{P_{g+1}+1} < V_g^{P_{g+1}} then
        for i = P_{g+1} + 1 to σ_{g+1} do
            P_{g+1} = P_{g+1} + 1;
            if V_g^i < V_g^{P_{g+1}} then
                P_{g+1} = i;
            end
        end
    end
end
```

each band, we design the local density in intragroup. Next, our main work is to acquire the threshold for each band. First, the similarity matrix corresponding to the bands in each group is chosen and arranged in ascending order to form matrix A. For the X_i in Q_g, the threshold θ_i is

$$\theta_i = \frac{\sum_{j=\lambda_g}^{P_{g+1}-1} A_{ij}}{P_{g+1} - \lambda_g} \tag{5}$$

where λ_g is the threshold parameter in Q_g. After obtaining the threshold θ_i, we count the number of the values that are nonzero and less than the threshold θ_i in the ith column of matrix A. Furthermore, the local density in intragroup δ_i is calculated by this number and the band numbers in Q_g.

The ranking strategy only focuses on intragroup information, which leads to high redundancy. To address the problem, we design the DIG, which can completely reflect the redundant for each band and other intergroup bands. First, the band with the maximum local density in intragroup is chosen to represent the main characteristic of each group. Subsequently, the difference r_i between intergroups is the average Euclidean distance for X_i and these bands. It should be noted that the representative band in the same group as X_i does not participate in the corresponding calculation.

Band selection aims to choose a band subset with rich content and low redundancy. Local density in intragroup and DIG can only be checked for the redundancy in bands. Considering this issue, we introduce the information entropy, a statistic that describes the degree of band disorder.

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The information entropy H_i is

$$H_i = -\sum_{i \epsilon L} \rho(X_i) \times \log(\rho(X_i)) \tag{6}$$

where $\rho(X_i)$ is the probability of X_i appearing in the image pixels. After obtaining these three factors, the evaluation criterion ζ_i is constructed to judge the significance in X_i, i.e.,

$$\zeta_i = \delta_i \times r_i \times H_i. \tag{7}$$

The band with the maximum value of evaluation criterion in each group is chosen to establish the band subset K, i.e.,

$$K = \left\{K_g\right\}_{g=1}^{g=m} \tag{8}$$

where K_g is the index of band with the maximum value in Q_g.

C. Optimal Number of Bands

Theoretically, the knowledge involved becomes richer as the number of bands increases. Nevertheless, the redundancy in band subset also increases. The optimal number of bands is a trade-off between superior performance and a small number of bands. The traditional methods [37], [41] adopt the classifier to acquire the optimal number of bands. The corresponding curve is generated by various band subsets. The inflection point is regarded as the optimal number.

These curve-based methods are susceptible to the influence of classifiers. To avoid using the classifier, an evaluation function is designed to replace the classifier. Specifically, the evaluation function adopts information content and redundancy to measure the performance in band subset K. The evaluation function χ_m is designed as follows:

$$\chi_m = \frac{\sum_{g=1}^{m} H_{K_g}}{m} \times \frac{\sum_{g=1}^{m-1} S_{K_g, K_{g+1}}}{m-1} \tag{9}$$

where the first term and the second term represent the information richness and the redundancy in K. We can produce the function value with various band subsets by (9). The curve is drawn by according to the number of bands and the corresponding function values. The inflection point is the point where the value of the function changes the most. The change ν_m is designed by the following equation:

$$\nu_m = \chi_m - \chi_{m-1}. \tag{10}$$

After generating different number of band subsets, the value is produced by (10). The optimal number of bands is the amount of subsets with the maximum value.

IV. Experiments

To verify the effectiveness of the proposed method, comparative experiments are conducted on various public hyperspectral datasets. First, three datasets are described in detail. Afterward, the classifiers and comparison methods are detailed. Finally, experiments from various aspects are discussed and analyzed.

A. Datasets

1) Pavia University: This dataset was collected by the reflective optics system imaging system (ROSIS) Sensor of Pavia University, Italy. The scene has 115 bands and each band contains 610 × 340 pixels. Since 12 bands have low SNR or water absorption, we only employ the remaining 103 bands for the experiment.

2) Salinas: This dataset was caught by the AVIRIS sensor in Salinas Valley, California, USA. There are 16 categories labeled in this scene. Each band has 512 × 217 pixels. Similarly, this scene removes 20 abnormal bands and remains 204 bands for analysis.

3) Botswana: The dataset was acquired by the NASA EO-1 satellite over the Okavango Delta in Botswana. The image includes 14 recognized categories. Besides, each band has 1476 × 256 pixels. After removing the noise and water absorption bands, 145 bands are available for study.

B. Experimental Setup

1) Classifier Setting: The classification of the obtained band subset is the main evaluation model for band selection. In the experiment, k-nearest neighbor (KNN) and support vector machine (SVM) classifiers are adopted to classify three datasets. To ensure the experimental accuracy, the same experimental parameters are set in the two classifiers. The parameter k of KNN is 5. The penalty factor and gamma coefficient of SVM are 1×10^4 and 0.5. In addition, 10% samples are randomly selected as the training set. To obtain stable results, we test each method five times and take the average as the final result. The overall accuracy (OA) and average OA (AOA) in bands as evaluation metrics to analyze performance. The number of bands is usually not known when the bands are chosen. We set the range is [3, 30] to discuss the performance for various methods.

2) Comparison Methods: Five methods are applied and compared with the proposed approach. Their main steps are described as follows.

1) MVPCA [34] establishes loading factor matrix. The matrix is utilized to rank the bands. The maximum ranked bands are selected to establish band subset.
2) E-FDPC [33] adopts local density and intraclass distance to rank the bands. The maximum ranked band in each cluster is chosen.
3) OPBS [36] ranks all the bands according to the variance and selects the bands with the maximum ranked as the initial subset. The final band subset is produced by updating the band subset and calculating the corresponding OP vector.
4) ASPS [39] acquires the partitioning points location by maximizing the ratio of intergroup distance to intragroup distance. The band with the minimum noise in each subspace constitutes the band subset.
5) FNGBS [40] applies coarse-fine strategy to divide bands into several groups. Furthermore, the local density and information entropy are employed to select one band in each group.

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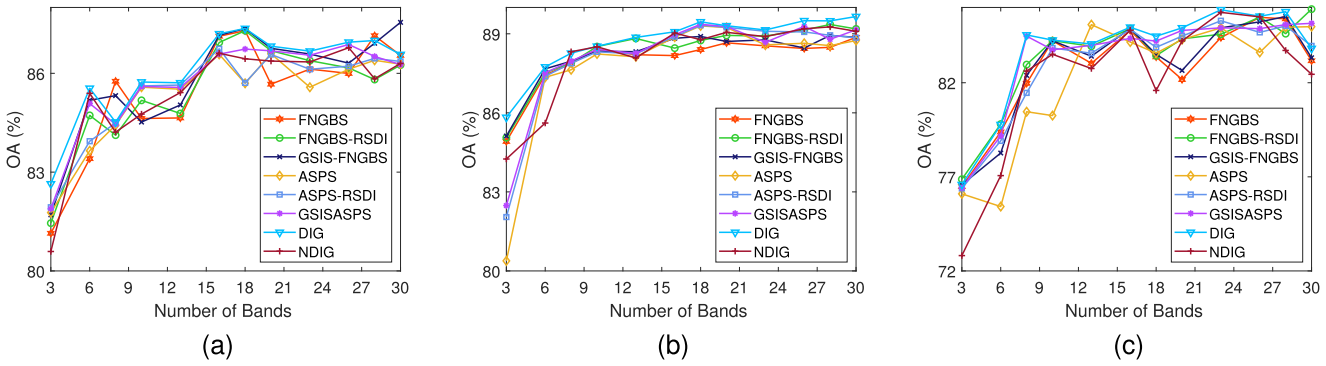


Fig. 3. OA in KNN classifier on three datasets. (a) Pavia University. (b) Salinas. (c) Botswana.

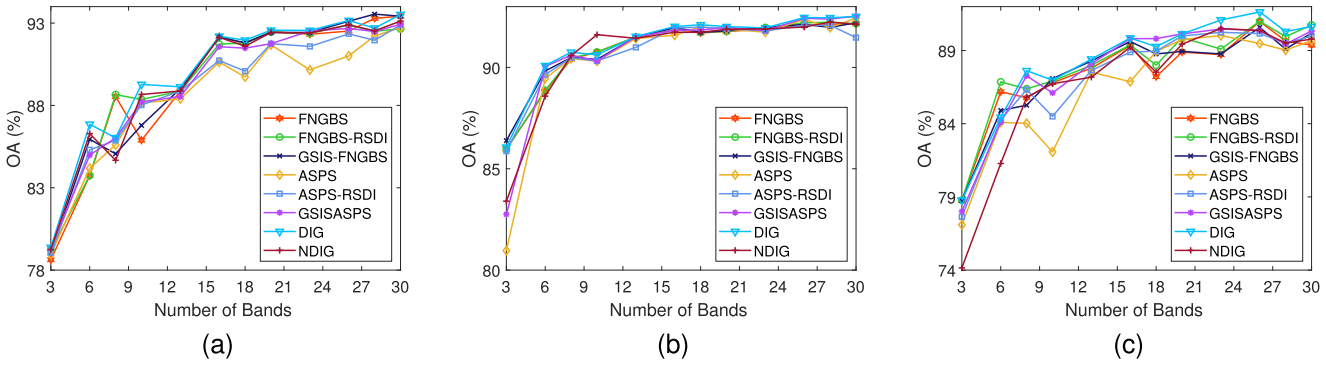


Fig. 4. OA in SVM classifier on three datasets. (a) Pavia University. (b) Salinas. (c) Botswana.

C. Ablation Study

The proposed DIG mainly includes GSIS and RSDI. To verify the validity of each part, two parts are respectively replaced with other approaches.

1) Effectiveness of the Grouping Strategy: Similar to the structure of DIG, FNGBS and ASPS combine grouping and ranking strategies. The grouping strategy in these two ways is replaced with GSIS. These approaches are named GSIS-FNGBS and GSIS-ASPS. To distinctly show superior performance in the proposed GSIS, the OA curves in GSIS-FNGBS, GSIS-ASPS, FNGBS, and ASPS by applying two classifiers are shown in Figs. 3 and 4.

The OA curve and AOA in GSIS-FNGBS has great advantages over FNGBS on three public datasets, as shown in Table I. Specifically, the grouping strategy in FNGBS only considers the similarity between the group center and the undetermined band. The effects of other intragroup bands are neglected. Similarly, the OA curve and AOA are higher in GSIS-ASPS than in ASPS. Particularly, when three bands are selected on Salinas dataset, the GSIS-ASPS in AOA is higher 2% and 5% than ASPS with the two classifiers. The grouping strategy in ASPS aims to maximize the ratio of intergroup and intragroup distances. However, the intergroup distance only considers the bands with the largest difference between the diverse groups, which is susceptible to noise bands. These results fully validate the superior capacity of GSIS.

2) Effectiveness of the Ranking Strategy: Similar, the ranking strategy in ASPS and FNGBS is replaced by RSDI to verify the effect of the proposed ranking strategy. These methods are referred to as ASPS-RSDI and FNGBS-RSDI. In addition, the proposed intergroup difference information is the major contribution that distinguishes DIG significantly from other methods. To verify the outstanding performance in this part, we design an approach that the DIG is not employed in the ranking strategy, which is named no difference between intergroups (NDIG).

In Figs. 3 and 4, compared with the original algorithms, the classification results in ASPS-RSDI and FNGBS-RSDI are greatly improved in performance. ASPS selects the band with the minimum noise within each group. FNGBS adopts information entropy and local density to choose the band with the largest product in each group. However, these two methods choose each band only considering its representation in a single group and ignoring its differences from other groups, which may lead to some redundancy in the band subset. In contrast to DIG, the OA in NDIG is considerably reduced when a small number of bands are generated. The gap in OA between DIG and NDIG on the SVM classifier even reaches 4.6% when three bands are selected on the Botswana dataset. The ranking strategy of NDIG only adopts the local density in intragroup and information entropy, which does not utilize intergroup difference information. The redundant information between bands from different groups is ignored in NDIG. Therefore, the performance of the band subset is degraded. In Table I, DIG achieves the best AOA on three public datasets. The above experiments not only verify the effectiveness of the proposed RSDI, but also demonstrate the outstanding contribution of introducing intergroup information to reduce the redundant data.

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TABLE I
AOA IN TWO CLASSIFIERS ON THREE DATASETS (%)

Dataset	Classifier	FNGBS [40]	FNGBS-RSDI	GSIS-FNGBS	ASPS [39]	ASPS-RSDI	GSIS-ASPS	DIG	NDIG
Pavia University	*K*NN	88.03	88.38	88.24	85.35	85.48	85.73	88.74	88.17
	SVM	90.92	90.97	91.03	88.62	89.03	89.38	91.21	90.77
Salinas	*K*NN	85.46	85.48	85.87	87.80	88.11	88.18	86.06	85.75
	SVM	89.47	89.60	89.56	90.55	90.88	90.84	89.94	89.57
Botswana	*K*NN	82.99	83.42	82.91	82.32	83.03	83.43	83.71	82.23
	SVM	87.46	87.92	87.52	86.55	87.34	87.83	88.26	86.79

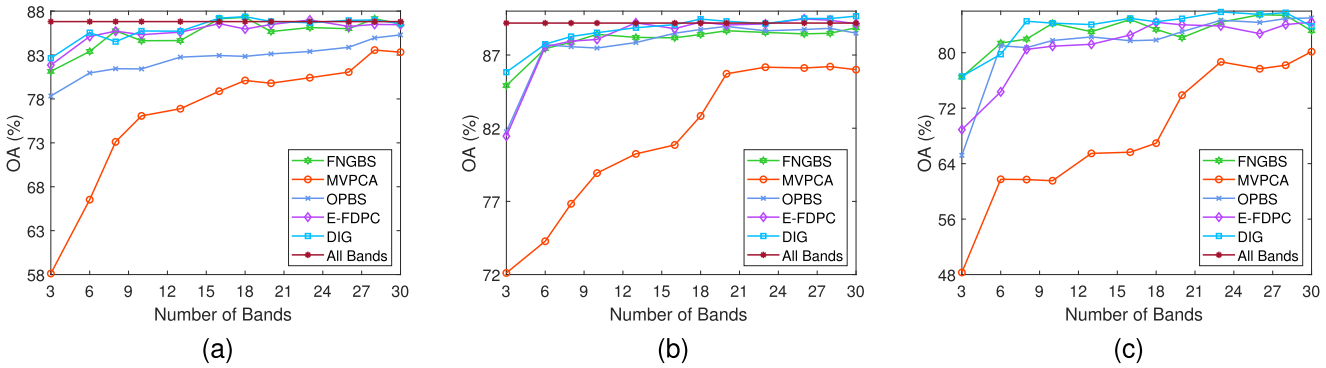


Fig. 5. OA in KNN classifier on three datasets. (a) Pavia University. (b) Salinas. (c) Botswana.

D. Comparison of Classification Performance

In this section, E-FDPC, FNGBS, MVPCA, and OPBS are compared with our DIG on three public datasets. Figs. 5 and 6 display the OA curves in five methods using KNN and SVM classifiers on three datasets. The OA curve in DIG is superior to the other algorithms. Moreover, the KNN classification accuracy in bands produced by DIG even exceeds that of all bands on Pavia University and Salinas datasets. Although FNGBS has a similar structure to the DIG, the OA curve in FNGBS is weaker than in DIG. There are two reasons for this consequence. First, the grouping strategy in FNGBS only takes into account the similarity between the group center and the bands near the group boundary, ignoring the influence of other intragroup bands. Besides, FNGBS neglects the possible redundancy in diverse groups bands when building band subset. Compared to DIG, the performance in OPBS and E-FDPC is not ideal. OPBS updates band subset based on OP vector, which only considers the redundancy between bands but ignores the information content. Contrary to OPBS, E-FDPC only focuses on the information content, which leads to large redundancy in the band subset. With regard to MVPCA, it simply ranks and chooses the bands with the maximum ranked. The redundant information in the bands is ignored. Consequently, it performs poorly on three public datasets.

Table II shows the AOA results for these five methods. Similar to the results for the OA curve, DIG exhibits superior performance in AOA. Besides, the gap between DIG and MVPCA reaches 9.5% by utilizing KNN on Pavia University dataset. Overall, DIG achieves the best performance on three public datasets. These data reveal from the side that DIG can obtain band subsets with rich information and low redundancy, which is consistent with the purpose of band selection.

TABLE II
AOA IN TWO CLASSIFIER ON THREE DATASET (%). THE BEST AOA IN DIVERSE CLASSIFIERS ON EACH DATASET IS SHOWN IN BOLD

Dataset	Method	*K*NN	SVM
Pavia University	FNGBS [40]	88.03	90.92
	MVPCA [34]	81.36	84.22
	OPBS [36]	87.68	90.72
	E-FDPC [33]	88.20	90.89
	DIG	**88.74**	**91.21**
Salinas	FNGBS [40]	85.46	89.47
	MVPCA [34]	76.49	79.55
	OPBS [36]	82.61	88.91
	E-FDPC [33]	85.73	89.39
	DIG	**86.06**	**89.94**
Botswana	FNGBS [40]	82.99	87.46
	MVPCA [34]	68.33	77.36
	OPBS [36]	81.38	86.48
	E-FDPC [33]	80.99	84.32
	DIG	**83.71**	**88.26**

E. Optimal Number Bands Comparison

According to the proposed method, we obtain the optimal number of bands on three datasets. The optimal number on Pavia University, Salinas and Botswana is 16, 22, and 21, respectively. The corresponding number of band subsets by

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TABLE III
SELECTED BANDS BY DIFFERENT METHODS IN OPTIMAL NUMBER. OPTIMAL NUMBER IS INDICATED IN PARENTHESES

Dataset	Method	Selected Bands
Pavia University(16)	FNGBS [40]	6, 12, 19, 23, 29, 35, 41, 48, 56, 62, 66, 76, 82, 88, 93, 99
	OPBS [36]	1, 3, 6, 23, 31, 38, 45, 66, 70, 72, 74, 76, 78, 81, 91, 102
	E-FDPC [33]	19, 33, 37, 42, 48, 50, 52, 54, 56, 61, 82, 86, 90, 92, 99, 101
	MVPCA [34]	53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68
	DIG	4, 11, 13, 20, 26, 32, 39, 45, 52, 58, 64, 71, 77, 83, 90, 102
Salinas(22)	FNGBS [40]	6, 16, 24, 32, 46, 54, 62, 68, 76, 90, 95, 113, 121, 126, 132, 141, 159, 165, 176, 185, 194, 196
	OPBS [36]	7, 11, 20, 29, 38, 40, 44, 49, 56, 59, 61, 69, 78, 79, 82, 86, 103, 126, 149, 165, 177, 190
	E-FDPC [33]	7, 11, 14, 20, 28, 32, 46, 54, 62, 69, 75, 83, 89, 94, 96, 97, 128, 141, 163, 177, 191, 199
	MVPCA [34]	40, 41, 42, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 68, 70, 71, 72
	DIG	9, 10, 19, 28, 37, 49, 56, 65, 74, 84, 93, 102, 116, 124, 130, 139, 158, 162, 167, 176, 185, 194
Botswana(21)	FNGBS [40]	5, 12, 21, 26, 31, 42, 43, 53, 58, 65, 70, 82, 86, 96, 100, 107, 112, 121, 129, 137, 140
	OPBS [36]	4, 6, 12, 18, 25, 27, 29, 32, 34, 42, 45, 48, 52, 56, 64, 72, 77, 92, 107, 123, 144
	E-FDPC [33]	11, 20, 42, 53, 63, 65, 67, 71, 72, 90, 91, 92, 94, 97, 99, 104, 106, 125, 126, 134, 140
	MVPCA [34]	7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27
	DIG	4, 7, 14, 21, 28, 35, 42, 49, 55, 68, 69, 76, 83, 89, 97, 103, 110, 123, 124, 131, 138

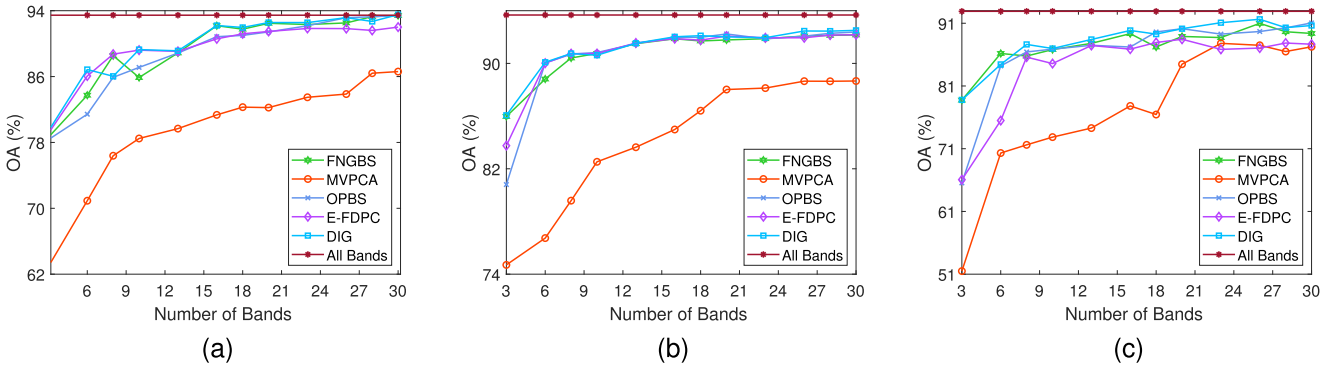


Fig. 6. OA in SVM classifier on three datasets. (a) Pavia University. (b) Salinas. (c) Botswana.

TABLE IV
OA FOR SELECTED BANDS (%). THE BEST AOA IN DIVERSE CLASSIFIERS ON EACH DATASET IS SHOWN IN BOLD

Dataset	Method	*K*NN	SVM
Pavia University	FNGBS [40]	87.13	92.18
	OPBS [36]	82.95	90.85
	E-FDPC [33]	86.57	90.58
	MVPCA [34]	78.88	81.35
	DIG	**87.20**	**92.20**
Salinas	FNGBS [40]	88.92	92.14
	OPBS [36]	89.21	92.11
	E-FDPC [33]	88.95	92.04
	MVPCA [34]	86.03	88.17
	DIG	**89.27**	**92.3**
Botswana	FNGBS [40]	84.16	88.55
	OPBS [36]	84.57	89.78
	E-FDPC [33]	84.02	86.49
	MVPCA [34]	76.41	87.00
	DIG	**85.26**	**90.2**

five band selection methods are respectively obtained from the three datasets. Furthermore, for the convenience of comparison, the bands are arranged in ascending order in Table III. Generally, it is considered a bad method if the produced bands are in a tiny range. The bands selected by most of the methods cover essentially all bands. Only the bands chosen by MVPCA are continuous over a small range on three datasets. Table IV shows the comparison results of the bands on two classifiers. DIG has significant superiority on three public datasets. In summary, the bands chosen by DIG are more dispersed and have better classification performance.

V. CONCLUSION

The article proposes a novel band selection method from the perspective of DIG. First, the similarity of intragroup is designed to reasonably partition the bands into multiple groups. Subsequently, the local density is improved and the DIG of each band is designed to reduce the redundancy in band subset. Based on the above work, we design an evaluation function to obtain the optimal number of bands. Experiments on three datasets demonstrate the great performance advantage of our DIG over other methods. In future work, we plan to adopt the grouping strategy and ranking strategy from a global perspective, which are treated as two optimization functions to obtain the global optimal solution.

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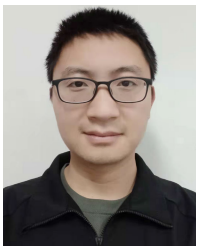
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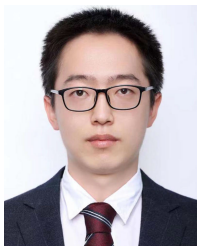
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